

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 19: Constructions (Circles)

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Circumscribed circle of a triangle

- Using ruler and compasses only, construct $\triangle ABC$ with $AB = 6$ cm, $AC = 4.5$ cm and $\angle BAC = 120^\circ$. Construct the circle that passes through A , B and C , and measure its radius.

ICSE 2025, 4 marks

SOLUTION

Draw $AB = 6$ cm. At A construct an angle of 120° (two 60° arcs) and cut off $AC = 4.5$ cm on its arm. Join BC .

Draw the perpendicular bisectors of AB and AC . Let them meet at O .

With centre O and radius OA , draw the circle. It passes through A , B and C .

Because $\angle A$ is obtuse, O lies outside the triangle, on the other side of BC from A .

Check by calculation: $BC^2 = 6^2 + 4.5^2 - 2(6)(4.5) \cos 120^\circ = 83.25$, so $BC \approx 9.12$ cm and the radius is $\frac{BC}{2 \sin 120^\circ} \approx 5.27$ cm.

Answer: the radius measures about 5.3 cm

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